Bayesian Wave Inversion and Uncertainty Quantification

Markov Chain Monte Carlo for Parameter Estimation in the Wave Equation

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June 5, 2026

1 Introduction

Waveform inversion aims to infer quantitative properties of an inaccessible medium from data collected by an array of sources and receivers that emit probing signals and record the resulting scattered waves. In practical settings, these measurements are inevitably contaminated by noise and subject to modeling uncertainties, which makes the inverse problem ill-posed and sensitive to perturbations in the data. To address this challenge, stochastic inference methods provide a natural framework for incorporating measurement uncertainty directly into the reconstruction process. In particular, Bayesian approaches allow the unknown parameters to be treated as random variables, enabling both stable inversion under noise and principled uncertainty quantification through posterior distributions. This work presents a numerical investigation of Bayesian waveform inversion using MCMC methods, focusing on how observation noise influences parameter recovery, posterior uncertainty, and identifiability in two representative inverse problems.

2 The Wave Model

To probe a medium with wave speed $c(\mathbf{x})$, a pulse $f(t)$ with finite support is emitted from a source located at $\mathbf{x}_s$. This generates a wave field $p^{(s)}(t,\mathbf{x})$ governed by

$$\left[\partial_t^2 - c^2(\mathbf{x})\Delta\right] p^{(s)}(t,\mathbf{x}) = f(t)\,\delta_{\mathbf{x}_s}(\mathbf{x}). \tag{1}$$

The wave operator can be rewritten as

$$\partial_t^2 + \mathcal{A}_c,$$

where the operator

$$\mathcal{A}_c = -\,c(\mathbf{x})\,\Delta\big(c(\mathbf{x})\cdot\big)$$

is positive definite and self-adjoint [1]. As a consequence, the wave measured at receiver locations $\mathbf{x}_r$, denoted by

$$\mathcal{M}_{r,s}(t) := p^{(s)}(t,\mathbf{x}_r), \qquad r,s = 1,\dots,m.$$

is transformed into a data matrix $\mathbf{D}(t)$ via

$$\mathbf{D}(t) = \mathcal{M}^f(t) + \mathcal{M}^f(-t),$$

where

$$\mathcal{M}^f(t) = -f'(-t) * \mathcal{M}(t), \qquad t > 0,$$

and $*$ denotes convolution in time. The entries of $\mathbf{D}(t)$ admit the representation

$$D_{r,s}(t) = \int_\Omega d\mathbf{x}\, u_0^{(r)}(\mathbf{x})\, u^{(s)}(t,\mathbf{x}),$$

where

$$u^{(s)}(t,\mathbf{x}) = \cos\big(t\sqrt{\mathcal{A}_c}\big)\, u_0^{(s)}(\mathbf{x}), \tag{2}$$

for $r, s = 1, \dots, m$. Here $u^{(s)}(t,\mathbf{x})$ solves the homogeneous wave equation

$$(\partial_t^2 + \mathcal{A}_c) u^{(s)}(t,\mathbf{x}) = 0, \qquad t > 0,\ \mathbf{x} \in \Omega, \tag{3}$$

with initial conditions

$$u^{(s)}(0,\mathbf{x}) = u_0^{(s)}(\mathbf{x}) = \widehat{f}\big(\sqrt{\mathcal{A}_c}\big)\, \delta_{\mathbf{x}_s}(\mathbf{x}), \qquad \partial_t u^{(s)}(0,\mathbf{x}) = 0.$$

Under these transformations, the forward map that relates the wave speed $c(\mathbf{x})$ to the scattered wave data can be expressed through a Lippmann–Schwinger-type integral equation:

$$D^{\text{sc}}_{r,s}(t) := D_{r,s}(t) - D_{r,s}(t; c_0) = \int_0^t dt' \int_\Omega d\mathbf{x}\, \rho(\mathbf{x})\, u^{(s)}(t',\mathbf{x})\, \partial_t u^{(r)}(t-t',\mathbf{x}; c_0), \tag{4}$$

where

$$\rho(\mathbf{x}) := \frac{c^2(\mathbf{x}) - c_0^2(\mathbf{x})}{c(\mathbf{x})\, c_0(\mathbf{x})}$$

denotes the contrast of the medium relative to the background wave speed $c_0(\mathbf{x})$.

The reconstruction of $c(\mathbf{x})$ from the scattered data $D^{\text{sc}}_{r,s}(t)$ is typically formulated as a PDE-constrained nonlinear least-squares problem, in which one seeks to minimize the data misfit. However, this approach is inherently challenging due to the strong nonlinearity of the forward mapping. Addressing this nonlinearity is beyond the scope of the present work; we refer the reader to the data-driven reduced-order model proposed in [1], which provides an effective linearization framework. Instead, we adopt the simplifying assumption that the internal wave field $u^{(s)}(t,\mathbf{x})$ is known. Under this assumption, the forward operator

$$A : \rho(\mathbf{x}) \mapsto D^{\text{sc}}_{r,s}(t)$$

becomes linear.

In practice, the measured data are corrupted by noise. In the next section, we introduce a stochastic model for this noise and discuss how Bayesian inference, combined with suitable prior assumptions, can be used to stabilize and regularize the inversion.

3 Bayesian Inference for Wave Speed Reconstruction

We consider a one-dimensional imaging setting in which the medium contrast $\rho(x)$ is parameterized by a finite-dimensional parameter vector

$$\theta = (\mu, \lambda) \in \Theta,$$

so that

$$\rho(x) = Q_{\mu,\lambda}(x). \tag{5}$$

The parameters (μ, λ) may represent, for example, the amplitude and position of a localized perturbation, or alternatively its amplitude and width. Their precise physical interpretation will be specified in the numerical experiments.

Let

$$D^{\text{sc}}(t; \theta)$$

denote the scattered wave field generated by the medium associated with the parameter vector θ. In practice, measurements are contaminated by noise, and we model the observed scattered wave data through the stochastic relation

$$X_t = X_0 \exp(D^{\mathrm{sc}}(t;\theta) + \sigma W_t)\,, \tag{6}$$

where $(W_t)_{t\geq 0}$ is a standard Brownian motion and $\sigma > 0$ controls the noise intensity. Given discrete observations

$$(X_{t_k})_{k=0}^{N}, \qquad t_k = k\Delta t,$$

we define the logarithmic observations

$$Y_{t_k} = \log X_{t_k}.$$

The increments

$$\Delta Y_k := Y_{t_{k+1}} - Y_{t_k}$$

then satisfy

$$\Delta Y_k = \Delta D_k^{\mathrm{sc}}(\theta) + \sigma\sqrt{\Delta t}\,\xi_k, \tag{7}$$

where

$$\Delta D_k^{\mathrm{sc}}(\theta) = D^{\mathrm{sc}}(t_{k+1};\theta) - D^{\mathrm{sc}}(t_k;\theta),$$

and $\xi_k \sim \mathcal{N}(0,1)$ are independent Gaussian random variables. Assuming independence of the increments, the likelihood function associated with the parameter vector $\theta = (\mu, \lambda)$ is given by

$$L(\theta) = \prod_{k=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma^2\Delta t}} \exp\left(-\frac{(\Delta Y_k - \Delta D_k^{\mathrm{sc}}(\theta))^2}{2\sigma^2\Delta t} \right). \tag{8}$$

For numerical computations it is more convenient to work with the log-likelihood,

$$\log L(\theta) = -\frac{N}{2}\log\bigl(2\pi\sigma^2\Delta t\bigr) - \frac{1}{2\sigma^2\Delta t}\sum_{k=0}^{N-1} (\Delta Y_k - \Delta D_k^{\mathrm{sc}}(\theta))^2\,. \tag{9}$$

In the Bayesian framework, the parameters are treated as random variables. Given a prior distribution $\pi(\theta)$ on the parameter space Θ, Bayes' theorem yields the posterior distribution

$$\pi(\theta \mid X) \propto L(\theta)\,\pi(\theta). \tag{10}$$

The posterior combines prior information with information extracted from the noisy wave measurements and provides a probabilistic characterization of the uncertainty associated with the inferred parameters. A classical inference method is the maximum likelihood estimator (MLE),

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta\in\Theta} L(\theta), \tag{11}$$

which seeks the parameter value that maximizes the likelihood of the observed data.

In general, the posterior distribution cannot be computed analytically due to the intractability of the normalization constant

$$Z = \int_{\Theta} L(\theta)\,\pi(\theta)\,d\theta.$$

Consequently, we employ Markov Chain Monte Carlo (MCMC) methods to generate samples from the posterior distribution using only the unnormalized density. These samples are then used both for parameter reconstruction and for uncertainty quantification of the inferred medium properties.

4 Metropolis–Hastings Algorithm and MCMC Sampling

Given the observed scattered wave data

$$\mathcal{D} = (X_{t_k})_{k=0}^{N},$$

we seek to sample from the posterior distribution

$$\pi(\theta \mid \mathcal{D}), \qquad \theta = (\mu, \lambda),$$

using the Metropolis–Hastings (MH) algorithm, a Markov Chain Monte Carlo (MCMC) method that constructs a Markov chain whose invariant distribution coincides with the target posterior distribution [4, 3, 6, 2].

Starting from an initial parameter vector

$$\theta^{(0)} = (\mu^{(0)}, \lambda^{(0)}),$$

the MH algorithm generates a sequence of samples by iteratively performing two steps:

1. propose a candidate parameter vector from a proposal distribution;
2. accept or reject the proposal according to an acceptance probability designed to preserve detailed balance.

Let

$$q(\theta' \mid \theta)$$

denote the proposal density. Given the current state

$$\theta = (\mu, \lambda),$$

a candidate state

$$\theta' = (\mu', \lambda')$$

is sampled according to

$$\theta' \sim q(\cdot \mid \theta).$$

The proposal is accepted with probability

$$\alpha(\theta, \theta') = \min\left(1, \frac{\pi(\theta' \mid \mathcal{D})\, q(\theta \mid \theta')}{\pi(\theta \mid \mathcal{D})\, q(\theta' \mid \theta)}\right). \tag{12}$$

If the proposal is rejected, the chain remains at the current state.

In this work, we employ a Gaussian random-walk proposal of the form

$$\mu' = \mu + \varepsilon_\mu, \qquad \varepsilon_\mu \sim \mathcal{N}(0, s_\mu^2), \tag{13}$$

$$\lambda' = \lambda + \varepsilon_\lambda, \qquad \varepsilon_\lambda \sim \mathcal{N}(0, s_\lambda^2). \tag{14}$$

Equivalently, letting

$$\theta = (\mu, \lambda),$$

the proposal kernel can be written as

$$q(\theta' \mid \theta) = \mathcal{N}\left(\theta, \begin{pmatrix} s_\mu^2 & 0 \\ 0 & s_\lambda^2 \end{pmatrix}\right).$$

The diagonal covariance structure corresponds to independent perturbations of the parameters, while the proposal variances s_μ^2 and s_λ^2 are tuned to obtain suitable acceptance rates and efficient exploration of the posterior distribution [6].

Under standard assumptions such as irreducibility, aperiodicity, and detailed balance, the resulting Markov chain is ergodic and converges in distribution toward the target posterior,

$$\theta^{(n)} \xrightarrow{d} \pi(\theta \mid \mathcal{D}), \qquad n \to \infty,$$

see [7, 5]. Consequently, empirical averages computed along the chain converge to posterior expectations, allowing numerical estimation of quantities such as posterior means, variances, credible intervals, and uncertainty bounds associated with the reconstructed medium parameters.

5 Numerical Experiments

Synthetic data are generated by first constructing the medium-dependent contrast $\rho(x)$ on a spatial grid. The resulting field is then propagated through the discretized wave operator to obtain the scattered wave signal. In matrix form, this is represented as

$$\mathbf{d}^{\mathrm{sc}} = A\rho,$$

where A denotes the discrete forward operator encoding the wave propagation and receiver sampling geometry. To account for measurement uncertainty, the observed data are modeled in log-space as a stochastic perturbation of the deterministic scattered field,

$$Y = \log X = d^{\mathrm{sc}} + \sigma W,$$

where W is a standard Brownian motion and $\sigma > 0$ controls the noise intensity. This induces the increment structure

$$\Delta Y_k = \Delta d_k^{\mathrm{sc}} + \sigma\sqrt{\Delta t}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0,1).$$

5.1 Inferring the amplitude and the width of a contrast profile

In this subsection we consider the inverse problem of reconstructing two medium parameters: the amplitude a and width w of a localized contrast profile. The medium is parameterized as

$$\rho(x) = Q_{a,w}(x) = a\exp\left(-\frac{(x-x_0)^4}{2w^2}\right), \tag{15}$$

where x_0 is a fixed reference location and (a, w) are the unknown parameters to be inferred.

Under this model, the likelihood function used for inference is implemented via the forward map

$$(a, w) \mapsto d^{\mathrm{sc}}(a, w) = A\rho(a, w),$$

where A is the discrete propagation operator. The log-likelihood is then computed from the mismatch between observed and predicted increments:

$$\log L(a, w) = -\frac{1}{2\sigma^2\Delta t}\sum_{k=0}^{N-1}\left(\Delta Y_k - \Delta d_k^{\mathrm{sc}}(a, w)\right)^2 + \text{const.} \tag{16}$$

Here, σ denotes the known noise level used in the data generation, and $\Delta d_k^{\mathrm{sc}}(a, w)$ is obtained from the discretized forward model $A\rho(a, w)$.

This formulation yields a two-dimensional Bayesian inverse problem in (a, w), where the goal is to infer the posterior distribution of the medium parameters from noisy scattered wave observations using the likelihood above combined with a chosen prior.

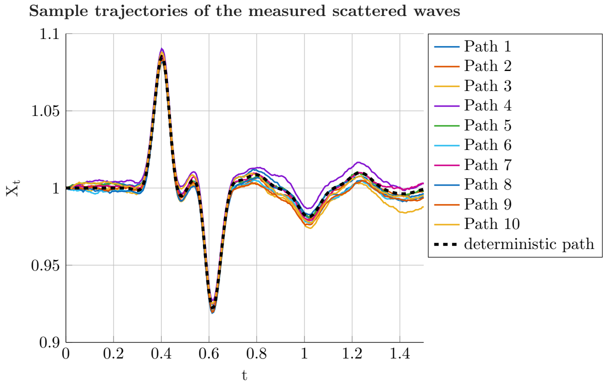


Figure 1: Sample trajectories of the stochastic scattered wave measurements generated from the deterministic scattered field $D^{\mathrm{sc}}(t)$ using the multiplicative noise model $X_t = X_0 \exp(D^{\mathrm{sc}}(t) + \sigma W_t)$, where $(W_t)_{t\geq 0}$ denotes standard Brownian motion. The noise intensity is set to $\sigma = 5 \times 10^{-3}$. The dashed curve corresponds to the deterministic scattered wave signal, while the solid curves represent independent noisy realizations used for Bayesian inference and uncertainty quantification.

The maximum likelihood estimates recover the true parameters with high accuracy despite the presence of measurement noise, indicating that the scattered wave data are informative enough for inference. In particular, the MLE yields $\hat{a}_{MLE} = 0.7489$ and $\hat{w}_{MLE} = 0.0050$, which are very close to the ground truth values $a = 0.75$ and $w = 0.0050$. This agreement suggests that the likelihood is sharply concentrated around the true parameter set under the given measurement time, sampling resolution and signal-to-noise ratio. Consequently, the inverse problem in this regime is well-conditioned at the statistical level, providing a strong foundation for subsequent Bayesian inference and uncertainty quantification.

Figure 2 shows the evolution of the Metropolis–Hastings chain and the corresponding posterior samples for the amplitude–width inference problem. The posterior rapidly concentrates near the ground truth parameters, and several samples become essentially indistinguishable from the exact values. The elongated structure in the joint posterior indicates a strong correlation between amplitude and width: increasing the contrast amplitude can be partially compensated by modifying the width, leading to similar scattered wave responses at the receivers. This coupling, also seen in Figure 3 reflects the entanglement between wave speed contrast intensity and effective propagation distance in the forward model. In contrast, this behavior is not observed in the subsequent amplitude–position inference experiment.

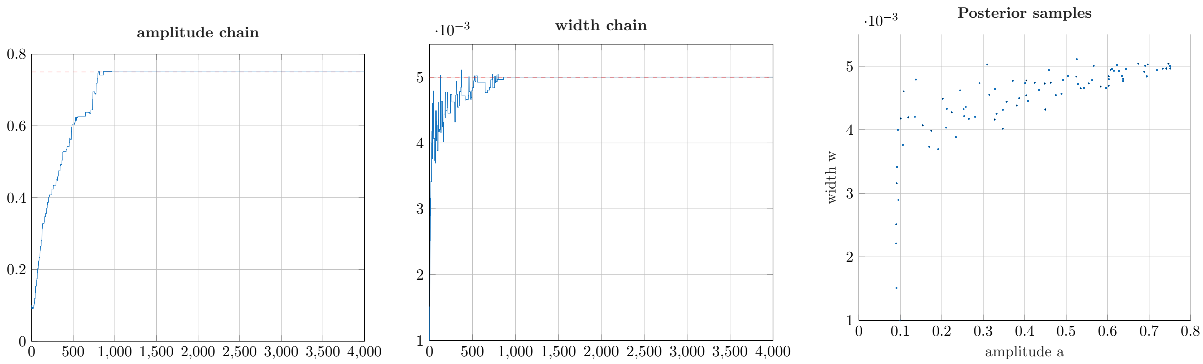


Figure 2: Metropolis–Hastings MCMC estimation of the contrast amplitude a and width w from noisy scattered wave measurements. The left and middle panels display the Markov chains for the amplitude and width parameters, respectively, while the right panel shows the corresponding posterior samples in parameter space. The proposal mechanism uses Gaussian random-walk updates with step sizes $s_a = 10^{-2}$ and $s_w = 10^{-3}$. A total of 4000 MCMC samples are generated. After a short transient phase, the chain concentrates around the true parameters $(a, w) = (0.75, 0.005)$, yielding posterior mean estimates $\hat{a} = 0.749$ and $\hat{w} = 0.005$.

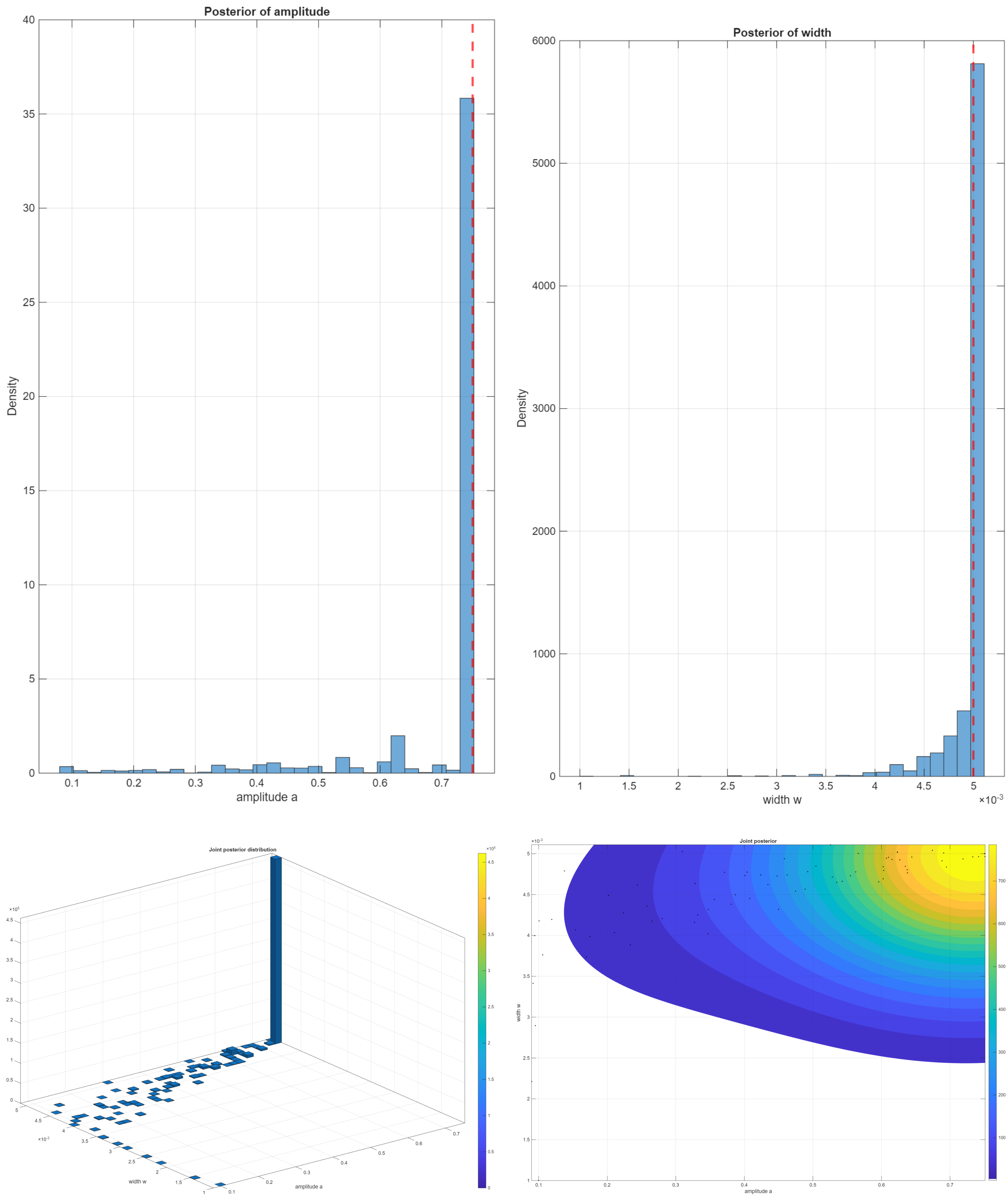


Figure 3: Posterior inference results for the amplitude–width parameter estimation problem based on 4000 MCMC samples. The top row shows the marginal posterior distributions for the amplitude a and width w, respectively, while the bottom row displays the joint posterior distribution obtained from histogram and smoothed kernel estimates. The isolines in the smooth joint posterior clearly reveal a strong negative correlation between the parameters: higher amplitudes must be compensated by smaller widths in order to reproduce the observed time-dependent scattered wave data. This reflects the intrinsic trade-off in the forward model, where different combinations of (a, w) can generate similar arrival-time signatures at the receivers.

Figure 4 shows the decay of the estimation error for the amplitude and width parameters as the number of MCMC samples increases, highlighting an initial $O(N^{-1/2})$ convergence followed by accelerated (superconvergent) behaviour once the chain concentrates in the high-probability region.

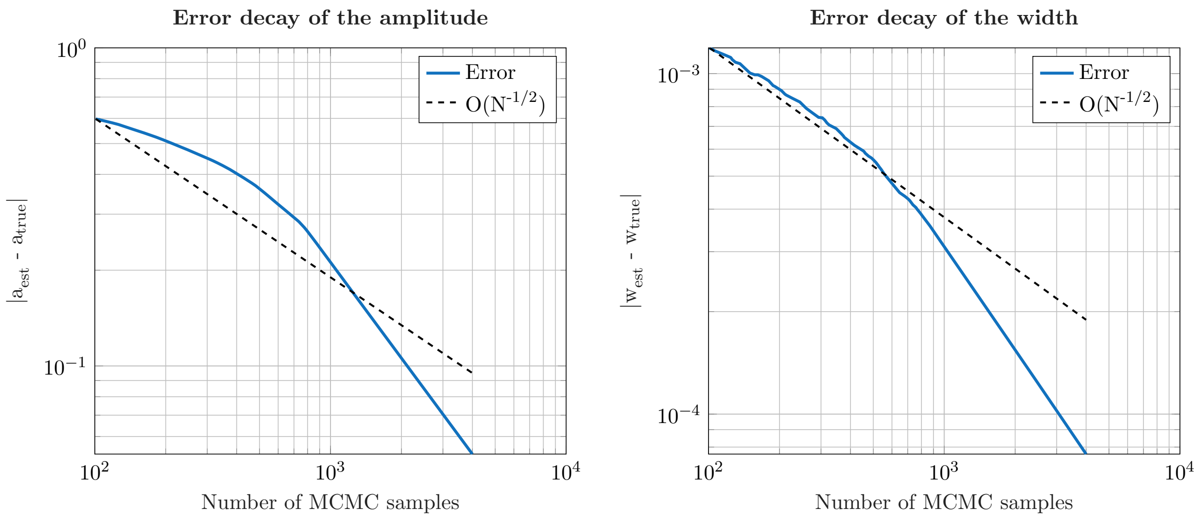
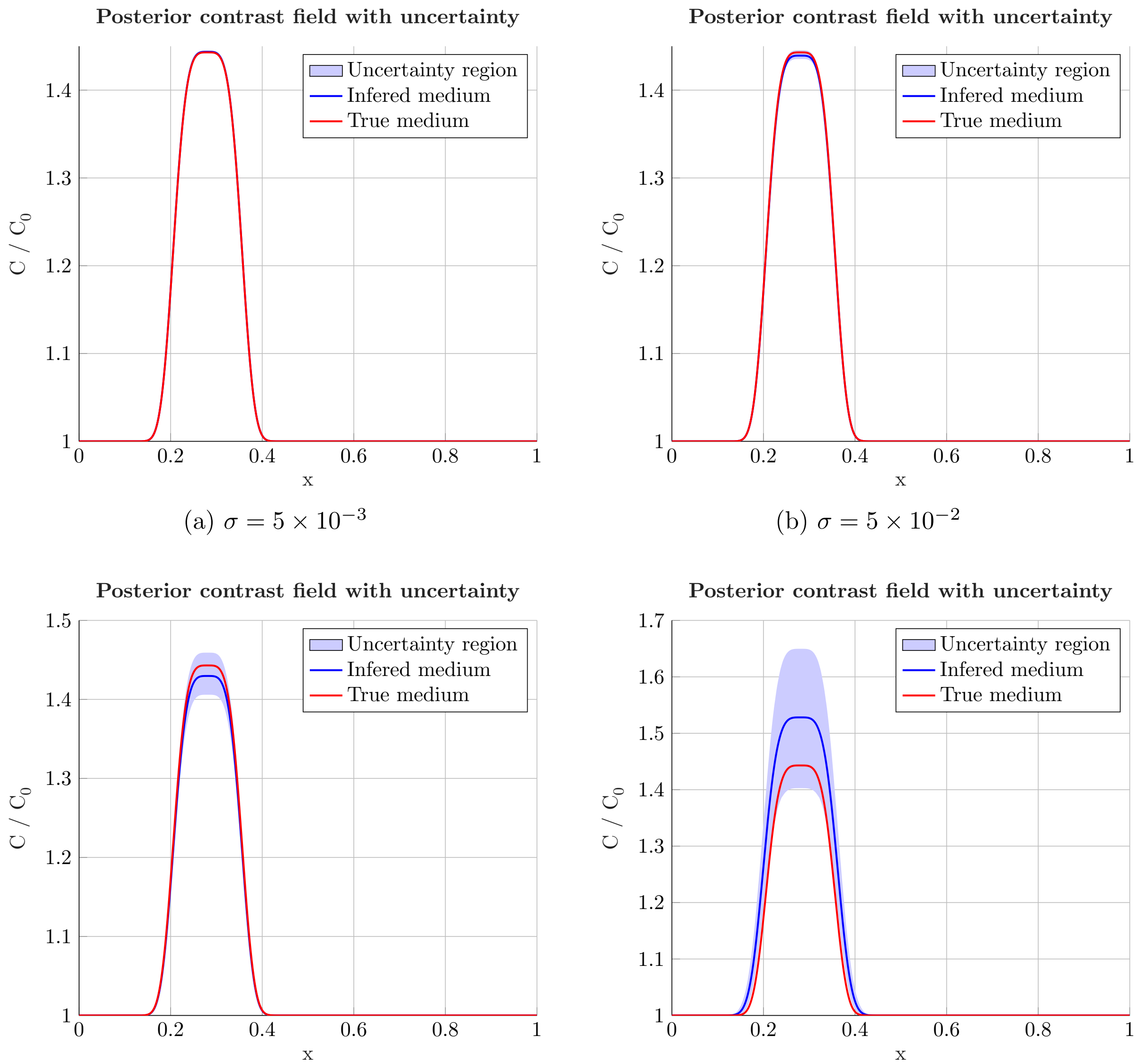


Figure 4: Decay of the estimation error for the amplitude a (left) and width w (right) as a function of the number of MCMC samples. The error in the width initially follows the expected Monte Carlo rate of decay, consistent with the law $O(N^{-1/2})$. For both parameters, a regime of accelerated convergence is observed after the initial transient phase, indicating a form of superconvergence once the chain has entered the high-probability region of the posterior. This behaviour reflects the increasing stability of the posterior mean estimates as the sampling concentrates around the dominant mode of the distribution.



(a) $\sigma = 5 \times 10^{-3}$

(b) $\sigma = 5 \times 10^{-2}$

(c) $\sigma = 2 \times 10^{-1}$

(d) $\sigma = 8 \times 10^{-1}$

Figure 5: Inferred medium and associated uncertainty regions for increasing noise levels $\sigma = 5 \times 10^{-3}$, 5×10^{-2}, 2×10^{-1}, 8×10^{-1}. The shaded uncertainty region is obtained via Monte Carlo propagation of posterior samples $\{(a_i, w_i)\}_{i=1}^{N_s}$ drawn after burn-in from the MCMC chain. Each sample generates a corresponding wave speed contrast $c^{(i)}(x)/c_0(x)$, from which pointwise credible intervals are computed as the empirical [2.5%, 97.5%] quantiles. Both uncertainty region and estimation error increase with the noise level, illustrating degradation of identifiability under stronger measurement noise.

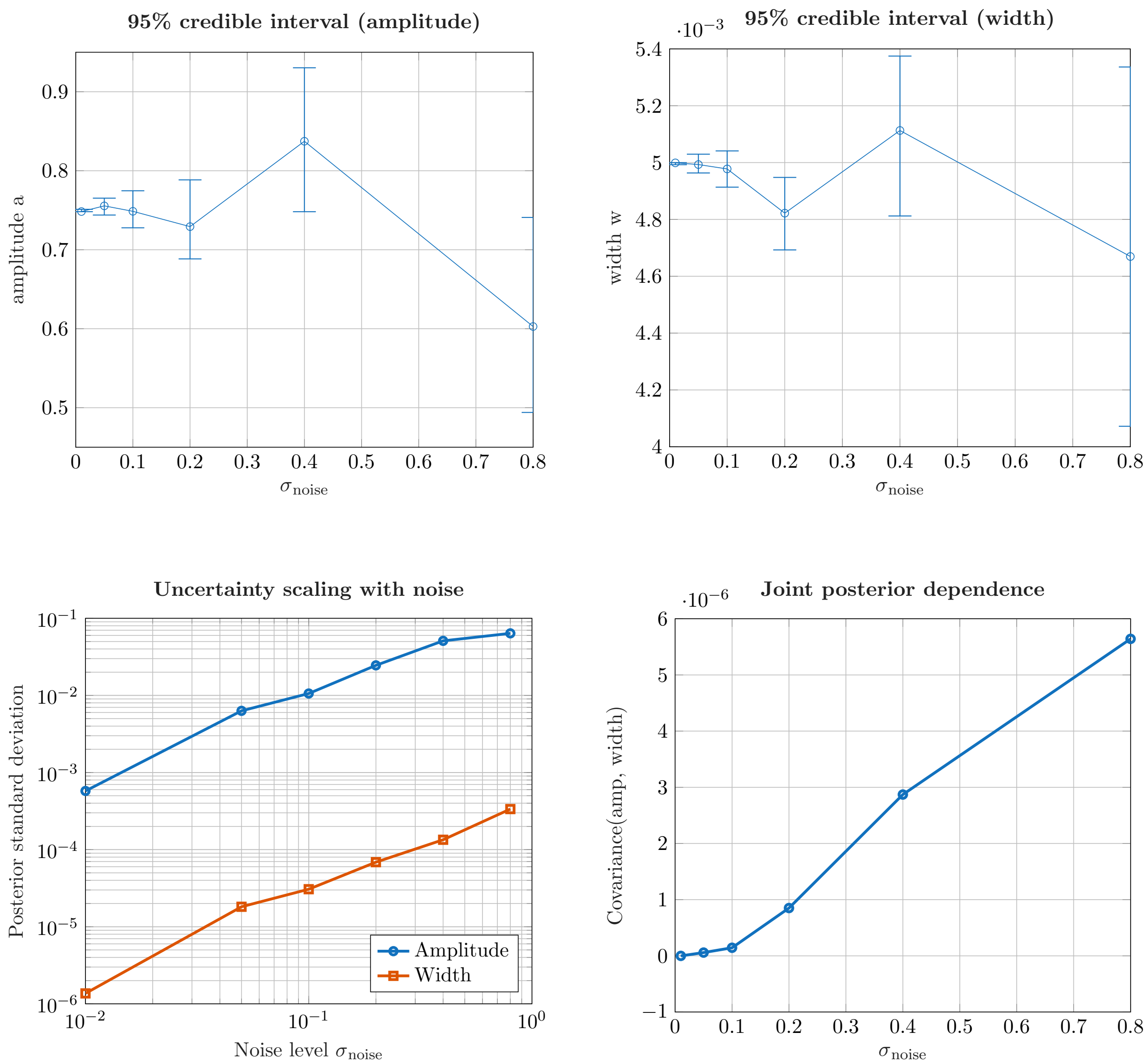


Figure 6: Posterior uncertainty analysis of wave parameter inference under increasing observation noise σ. The top row shows the evolution of the posterior mean and 95% credible intervals for the amplitude and width parameters, respectively. As noise increases, the credible intervals widen and the posterior means deviate further from the nominal values, indicating deteriorating estimation accuracy. The bottom-left panel shows the posterior standard deviation of both parameters, computed from MCMC samples after burn-in. Both variances increase monotonically with noise level, confirming the expected loss of information in the inverse problem as measurement noise increases. The bottom-right panel shows the posterior covariance between amplitude and width. The positive covariance indicates increasing statistical dependence (parameter entanglement) as noise increases, implying that the parameters become more strongly coupled under high-noise regimes.

5.2 Inferring the amplitude and position of a contrast profile

In this subsection we consider the inverse problem of reconstructing two medium parameters: the amplitude a and spatial position $x_0(p)$ of a localized contrast profile. The medium is parameterized as

$$\rho(x) = Q_{a,p}(x) = a \exp\left(-\frac{(x - x_0(p))^4}{2w^2} \right), \tag{17}$$

where w is a fixed width parameter and (a, p) are the unknown parameters to be inferred.

Figure 7 illustrates the increased difficulty of jointly inferring the amplitude and spatial position of the contrast profile compared to the amplitude–width inference problem. For the same noise level $\sigma_{\text{noise}} = 5 \times 10^{-3}$, a larger number of MCMC samples (8000 iterations) is required to obtain stable posterior estimates. In particular, the position chain exhibits occasional large jumps, reflecting the strong sensitivity of the scattered wave data to spatial misalignment of the contrast profile. Incorrect position proposals produce substantial arrival-time discrepancies in the observed wave field, leading to large variations in the data misfit functional. Such behavior was not observed in the width inference setting, where the position parameter was fixed and perturbations of the width induced comparatively weaker timing effects in the scattered signals.

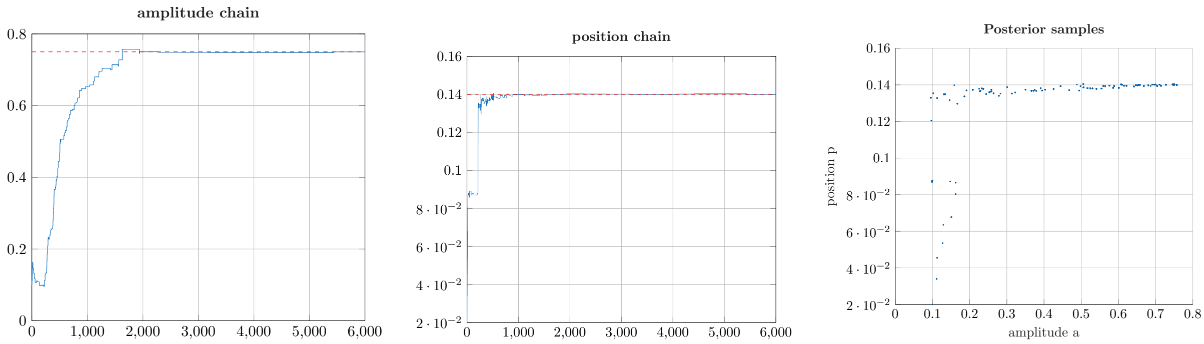
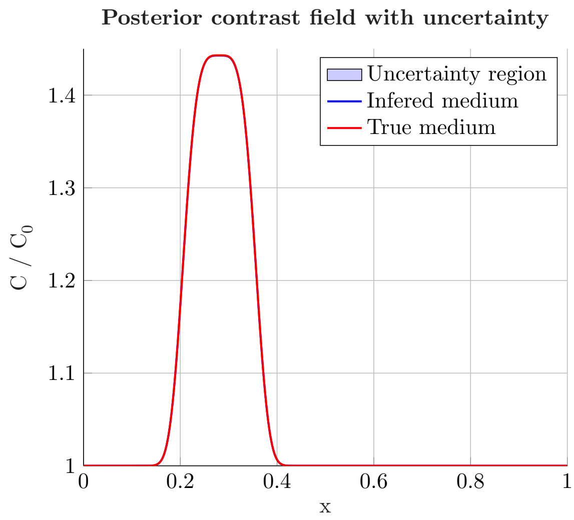
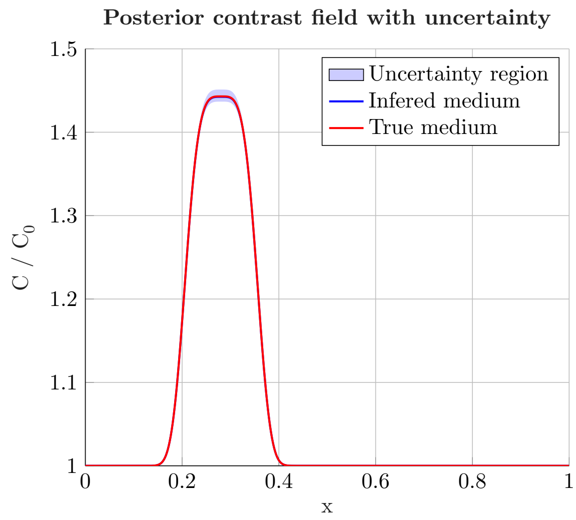


Figure 7: Metropolis–Hastings MCMC estimation of the contrast amplitude a and position $x_0(p)$ from noisy scattered wave measurements with noise level $\sigma_{\text{noise}} = 5 \times 10^{-3}$. The left and middle panels display the Markov chains for the amplitude and position parameters, respectively, while the right panel shows the corresponding posterior samples in parameter space. The proposal mechanism uses Gaussian random-walk updates, and a total of 8000 MCMC samples are generated. Despite the increased posterior complexity associated with position inference, the chain eventually concentrates around the true parameters and yields accurate posterior estimates.

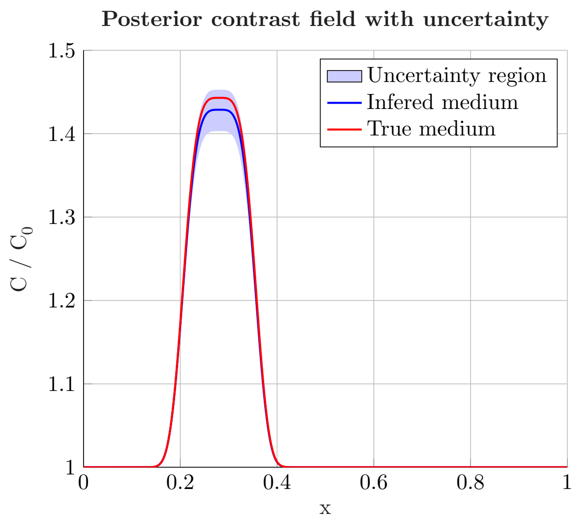
Figure 8 shows the inferred medium and corresponding uncertainty regions for increasing noise levels. As the noise level increases, the uncertainty bands become progressively wider, indicating reduced identifiability of the underlying parameters.



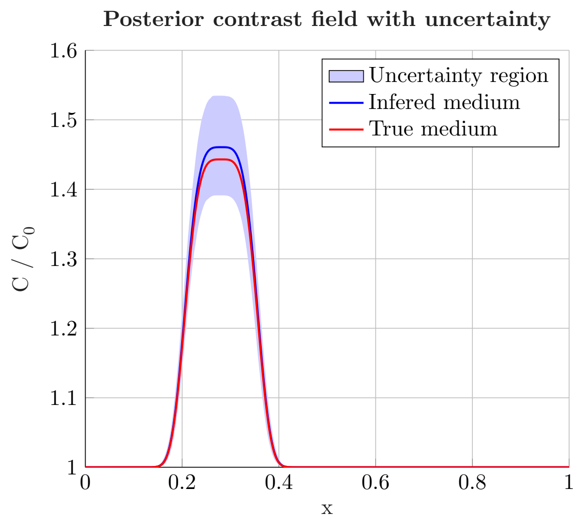
(a) $\sigma = 5 \times 10^{-3}$



(b) $\sigma = 5 \times 10^{-2}$



(c) $\sigma = 2 \times 10^{-1}$



(d) $\sigma = 8 \times 10^{-1}$

Figure 8: Inferred medium and associated uncertainty regions for increasing noise levels $\sigma = 5 \times 10^{-3}$, 5×10^{-2}, 2×10^{-1}, 8×10^{-1}. The shaded uncertainty region is obtained via Monte Carlo propagation of posterior samples $\{(a_i, p_i)\}_{i=1}^{N_s}$ drawn after burn-in from the MCMC chain. Each sample generates a corresponding wave speed contrast $c^{(i)}(x)/c_0(x)$, from which pointwise credible intervals are computed as empirical [2.5%, 97.5%] quantiles. The estimation error increases with noise level, illustrating degradation of identifiability under stronger measurement noise. The uncertainty region also increases with noise level and becomes more pronounced in amplitude than in spatial position, as additionally shown in Figure 9.

Figure 9 shows the expected growth of posterior uncertainty as the observation noise increases. The credible intervals widen progressively for both parameters, indicating increasing ambiguity in the inverse reconstruction problem. The amplitude estimates remain relatively stable for moderate noise levels but deteriorate significantly for $\sigma_{\text{noise}} = 0.8$, where both posterior spread and estimation bias become substantial. In contrast, the position parameter retains comparatively smaller absolute uncertainty, although its credible interval also grows steadily with noise. The posterior standard deviations increase monotonically with the noise level, with the amplitude exhibiting stronger sensitivity to observational perturbations than the position parameter. The positive posterior correlation between amplitude and position indicates statistical coupling between the parameters, suggesting that variations in amplitude can partially compensate for shifts in the inferred position.

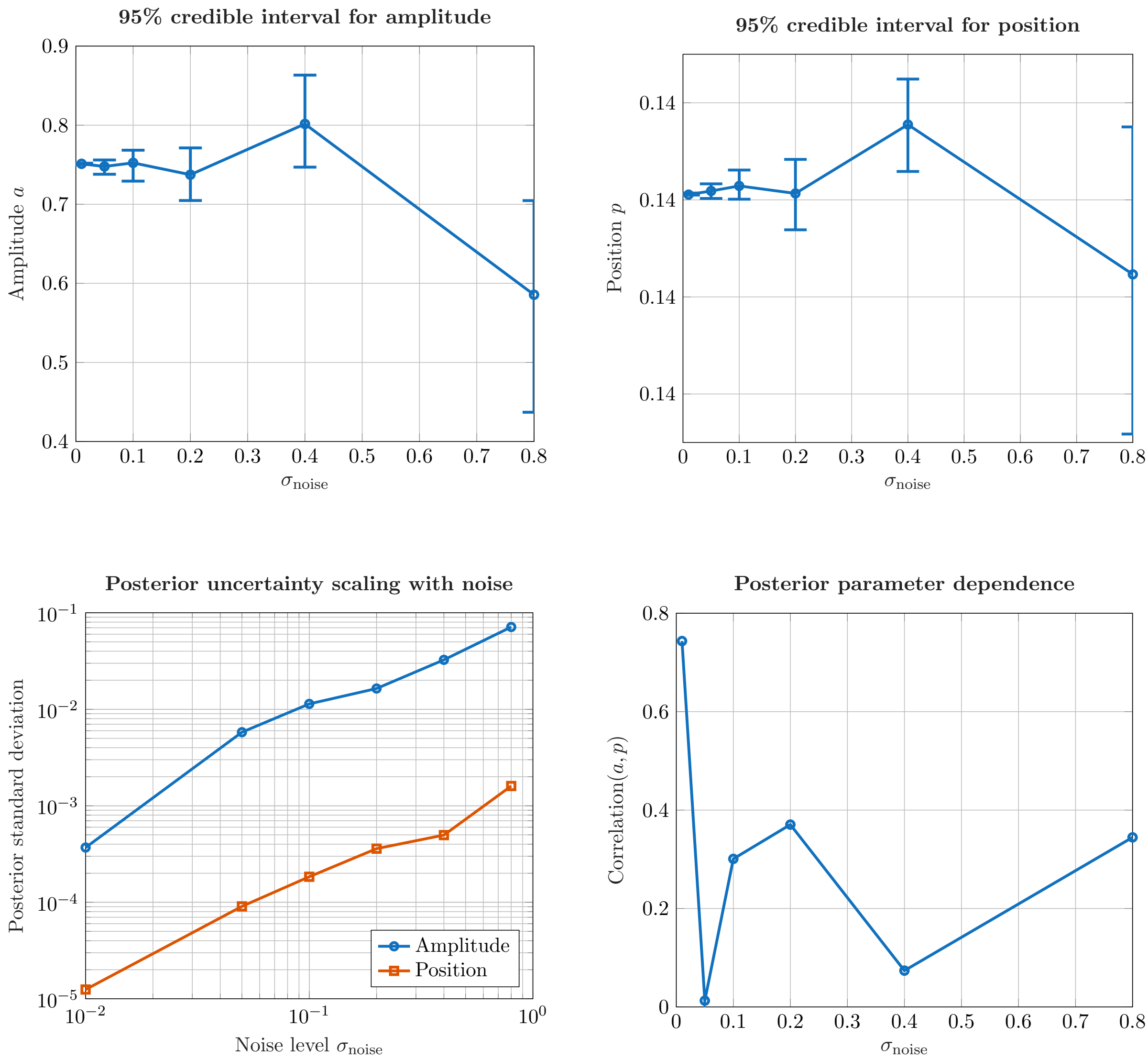


Figure 9: Posterior uncertainty analysis for the joint inference of the contrast amplitude a and position $x_0(p)$ under increasing observation noise σ_{noise}. The top row shows the posterior means and corresponding 95% credible intervals for the amplitude and position parameters, while the bottom-left panel displays the posterior standard deviations computed from MCMC samples after burn-in. The bottom-right panel shows the posterior correlation between amplitude and position. As the noise level increases, both posterior uncertainty and parameter coupling increase, leading to reduced reconstruction accuracy and deteriorating identifiability.

The two numerical experiments—(i) joint inference of amplitude and width, and (ii) joint inference of amplitude and position—consistently demonstrate how observation noise controls the quality of parameter reconstruction in the inverse wave problem. In both cases, increasing noise leads to broader posterior distributions, larger credible intervals, and reduced accuracy of the inferred parameters. This reflects the progressive loss of information in the data and the corresponding degradation of identifiability. A key difference between the two settings lies in the sensitivity of the parameters to the forward model. In the amplitude–position problem, the posterior exhibits stronger nonlinearity and occasional large MCMC transitions, since misalignment in position induces significant phase and arrival-time shifts in the wave field. In contrast, the amplitude–width problem shows smoother posterior geometry, as variations in width affect the data more gradually.

6 Conclusion

This study has presented a Bayesian framework for inverse wave problems using Metropolis–Hastings MCMC, applied to the joint inference of amplitude–width and amplitude–position parameterizations of a localized contrast profile. Across both numerical studies, the approach enables not only accurate parameter reconstruction under moderate noise levels, but also a rigorous quantification of uncertainty through posterior distributions.

The comparative study between the amplitude–width and amplitude–position cases further highlighted the role of parameterization in inverse problem conditioning. In particular, position inference exhibits stronger sensitivity to measurement perturbations due to its direct impact on wave arrival structure, leading to more pronounced posterior nonlinearity and stronger parameter coupling. In contrast, width variations induce smoother changes in the data, resulting in more stable posterior geometry. A possible direction for future investigation is the replacement of the standard (L^2)-based log-likelihood with a Wasserstein-type metric. Such an approach could help control the steep jumps observed in the MCMC position chain, since the Wasserstein distance naturally accounts for transport of the contrast profile when correcting amplitudes alone becomes too costly.

Overall, the results emphasize that Bayesian inference via MCMC provides a significant advantage over deterministic inversion methods by delivering not only point estimates, but also full uncertainty quantification. This probabilistic characterization is essential for assessing reliability, identifiability, and parameter coupling in ill-posed inverse problems, particularly under increasing observational noise.

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